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Step 4 - Counter Scaling

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture, specifically the System Counter and Counter Scaling functionality introduced in ARMv8.4-A. ARM allows the System Counter to advance by a configurable amount rather than always incrementing by one.

Learning Objectives

After completing this chapter, you should understand:

- What Counter Scaling is
- Why ARM introduced Counter Scaling
- Internal System Counter representation
- CNTSCR fields
- Integer and fractional increments
- Difference between Counter Scaling and PLLs
- How scaling affects the observed counter value

1. Traditional System Counter

Historically the System Counter behaved as:

Counter Tick	Counter Update
Tick 1	Count += 1
Tick 2	Count += 1
Tick 3	Count += 1

Result:

Tick Number	Count Value
0	0

1	1
2	2
3	3

One clock tick results in one counter increment.

2. Why Counter Scaling Was Introduced

ARM introduced Counter Scaling in Armv8.4-A. Instead of incrementing by one on every tick, the counter can increment by a software-configurable value. [support.arm.com]

Conceptually:

Mode	Counter Update
Traditional	Count += 1
Scaled	Count += X

where:

X = Scaling Value

3. System Counter Internal Representation

When Counter Scaling is supported, ARM internally expands the System Counter to:

Portion	Width
Integer Part	64 bits
Fractional Part	24 bits
Total	88 bits

The processor only sees:

64-bit Integer Portion

The:

24-bit Fractional Portion

is maintained internally by the System Counter. [support.arm.com]

4. Counter Scaling Register (CNTSCR)

The scaling value comes from:

CNTSCR

Counter Scaling Register.

Format:

Bits	Field
[31:24]	Integer Part
[23:0]	Fractional Part

Total Width:

32 bits

[support.arm.com]

5. CNTSCR Field Interpretation

Example 1

Field	Value
Integer	1
Fraction	0

Increment:

1.0

Counter behavior:

Tick	Count
0	0
1	1
2	2
3	3

Equivalent to traditional operation.

Example 2

Field	Value
Integer	2
Fraction	0

Increment:

2.0

Counter behavior:

Tick	Count
0	0
1	2
2	4
3	6

Counter appears to advance twice as fast.

6. Fractional Scaling

Suppose:

Field	Value
Integer	1
Fraction	0x800000

Fraction value:

0x800000 / 2^24
=
0.5

Total increment:

1.5

Internal behavior:

Tick	Internal Count
0	0.0
1	1.5
2	3.0
3	4.5
4	6.0

The fractional portion accumulates internally. [support.arm.com]

7. Counter Scaling vs PLL

These concepts are frequently confused.

Feature	PLL	Counter Scaling
Modifies Clock Frequency	Yes	No
Modifies Counter Increment	No	Yes
Creates New Clock	Yes	No
Changes Counter Math	No	Yes

PLL Example

24 MHz
↓
48 MHz

Clock changed.

Counter Scaling Example

24 MHz Clock
↓
Count += 2

Clock unchanged.

Counter behavior changed.

8. Why Fractional Scaling Exists

Without fractional scaling:

Possible increments:

1
2
3
4

With fractional scaling:

Possible increments:

1.25
1.5
1.75
2.125

This gives ARM more flexibility in how time is represented. [support.arm.com]

9. Architectural View

Component	Purpose
Counter Clock	Generates ticks
CNTSCR	Defines increment amount
System Counter	Accumulates count value
Integer Portion	Visible to software
Fractional Portion	Maintained internally

Key Takeaways

Concept	Key Point
Counter Scaling	Introduced in Armv8.4-A
Traditional Counter	Count += 1
Scaled Counter	Count += X
System Counter	Internally 88 bits
CNTSCR	8-bit integer + 24-bit fraction
Fractional Scaling	Supported
PLL	Changes clock frequency
Counter Scaling	Changes counter increment

Knowledge Check

Q1

What problem does Counter Scaling solve?

Q2

What is the width of the internal System Counter when scaling is enabled?

Q3

What are the two fields of CNTSCR?

Q4

What is the difference between Counter Scaling and a PLL?

Q5

Why is a fractional portion required?

Answer Key

Question	Answer
Q1	Allows configurable counter advancement without changing clock frequency
Q2	88 bits
Q3	8-bit integer field and 24-bit fractional field
Q4	PLL changes clock frequency, Counter Scaling changes counter increment amount
Q5	Allows non-integer count advancement